BIO220 – EXERCISES FOR COURSEWORK 1

1) Which of the following statements about the electric field $\boldsymbol{E}$ is correct?

A) The electric field changes with the velocity of the particle experiencing it.

B) The electric field is a vector field.

C) The electric field is null at the middle point of a positive charge +q and a negative charge -q.

D) None of the above answers are correct.

2) Which of the following statements about electrostatic energy is correct?

A) The electrostatic potential energy is conserved along any trajectory.

B) The electrostatic potential energy is a vector.

C) If a charge is moving in an electrostatic field the sum of electrostatic potential and kinetic energy does not change with time.

D) The formula for electrostatic energy is always $U = kQq/r$.

3) An electron with initial velocity parallel to the y axis: $\boldsymbol{v} = (0, v_y, 0)\ m/s$ in an electric field parallel to x axis: $\boldsymbol{E} = (E_x, 0,0)\ N/C$. The mass of the electron is m_e and its charge is $-e$.

What is the final velocity ($\boldsymbol{v_f}$) of the electron after an arbitrary time t?

A) $v_f = \left(\frac{-eE_x t}{m_e}, v_y, 0\right)$

B) $v_f = \left(0, v_y - \frac{eE_x t}{m_e}, 0\right)$

C) $v_f = \left(\frac{eE_x t}{m_e}, v_y, 0\right)$

D) $v_f = \sqrt{\left(\frac{eE_x t}{m_e}\right)^2 + v_y^2}$

4) Which of the following statements about molecular polarization is correct:

A) All molecules have intrinsic polarization.

B) All molecules are polarized by a sufficiently strong electric field.

C) Water is a natural dipole because the oxygen atom is more positive than the center of mass of the two hydrogen atoms.

D) Carbon dioxide is a natural dipole because the oxygen atoms are more negative than the carbon atom.

5) Which of the following statements about wave functions is correct?

A) The electron wave function is the probability density of finding an electron in a region of space.

B) The wave function can change (collapse) after a measurement is taken.

C) The wave function for a proton cannot be defined.

D) All the answers above are correct.

6) Which of the following statements about atomic and molecular orbitals is correct?

A) All molecules are stable at room temperature.

B) In the hydrogen atom, only the orbital 1S can be occupied.

C) In the hydrogen atom, the electron always has spin “up”.

D) Water is a natural dipole as the electrons pass more time around the oxygen atom compared to the two hydrogen atoms.

7） Which of the following statements about force fields is correct?

A) The Lennard-Jones potential is purely attractive at all distances.

B) Dihedral angles are always kept fixed in molecular simulations because rotation is not allowed.

C) In classical force fields, chemical bonds are typically modeled as harmonic oscillators.

D) All force field parameters must be derived exclusively from experimental data.

8） Which of the following statements about Ramachandran plots is correct?

A) All combinations of ϕ and ψ dihedral angles are equally probable in proteins.

B) Ramachandran plots show that certain ϕ and ψ angle combinations are forbidden due to steric

clashes.

C) Glycine and proline have identical Ramachandran plots.

D) The Ramachandran plot is only valid for nucleic acids, not proteins.

9) Which of the following statements about dihedral angles is correct?

A) Dihedral angles are defined by three consecutive atoms forming two bonds.

B) All dihedral angles have only one energy minimum corresponding to a single stable conformation.

C) In proteins, the ϕ and ψ dihedral angles describe the relative position of two consecutive amino acids.

D) Cis-trans isomerism cannot occur in dihedral angles because it requires breaking chemical bonds.

10) Which of the following statements about molecular orbitals is correct?

A) Molecular orbitals are stationary solutions to the Schoedinger equation for the electrons.

B) The energy required to excite electrons in a hydrogen molecule (H_2) at room temperature (310K) is comparable to thermal energy.

C) All molecular orbitals have spherical symmetry like the 1s atomic orbital.

D) Water has no dipole moment because its molecular orbitals are perfectly symmetric.